



JAVA using BlueJ



Values and Types

The Character set is the defined list of characters, recognized by the compiler or interpreter.

Letters	Digits	White Space
Capital A to Z	All decimal digits 0 to 9	Blank Space
Small a to z		Horizontal tab
		Vertical tab
		New Line

Special Character

,	Comma	&	Ampersand
.	Dot	^	Caret
;	Semicolon	*	Asterisk
:	Colon	-	Minus
'	Apostrophe	+	Plus
“	Quotation Mark	<	Less than
!	Exclamation mark	>	Greater than
	Vertical bar	()	Parenthesis
/	Slash	[]	Bracket
~	Tilde	%	Percent
_	Underscore	#	Hash
\$	Dollar	=	Equal to
?	Question mark	@	At the rate

Java tokens are the basic building blocks in java language which are constructed together to write a java program. Each and every smallest individual units in a java program are known as java tokens.

- Punctuators / Separators
- Keywords
- Identifiers
- Constants/Literals
- Operators
- Comments

Punctuators / Separators

Every language has a unique set of special symbols that work as punctuators / separators. In java, punctuators are used to grouping and separating the data values written in java programs.

Symbol	Name	Purpose
;	Semicolon	Terminate the statement
()	parenthesis	Used in expression and function
[]	Square bracket	Use for array declaration
{ }	Curly Brace	Scope of the statement
,	Comma	Variable separator
.	Period (dot)	to access the member variables and methods.

Keywords

Keywords are special words that have special meaning in programming language. They represent a specific task to be performed. Keywords cannot be used as identifiers. All keywords are used in lowercase.

<code>abstract</code>	<code>do</code>	<code>implements</code>	<code>private</code>	<code>throw</code>
<code>boolean</code>	<code>double</code>	<code>import</code>	<code>protected</code>	<code>throws</code>
<code>break</code>	<code>else</code>	<code>instanceof</code>	<code>public</code>	<code>transient</code>
<code>byte</code>	<code>extends</code>	<code>int</code>	<code>return</code>	<code>true</code>
<code>case</code>	<code>false</code>	<code>interface</code>	<code>short</code>	<code>try</code>
<code>catch</code>	<code>final</code>	<code>long</code>	<code>static</code>	<code>void</code>
<code>char</code>	<code>finally</code>	<code>native</code>	<code>super</code>	<code>volatile</code>
<code>class</code>	<code>float</code>	<code>new</code>	<code>switch</code>	<code>while</code>
<code>continue</code>	<code>for</code>	<code>null</code>	<code>synchronized</code>	
<code>default</code>	<code>if</code>	<code>package</code>	<code>this</code>	

All Java components require names. Names used for classes, variables, and methods are called identifiers.

- It can be only combination of alphabet (a-z, A-Z), digits (0-9) , under score) (_) and (dollar (\$) sign.
- All identifiers should begin with a letter (A to Z or a to z), dollar (\$) or an underscore (_).
- A keyword cannot be used as an identifier.
- identifiers are case sensitive and have no maximum length.

Identify the valid Identifiers

1. simple_interest
2. Simple interest
3. 123num
4. num_123
5. num123
6. num@123
7. While
8. ELSE
9. True
10. NUMBER

Identify the valid Identifiers

1. **simple_interest**
2. **Simple interest** # contain the space
3. **123num** # start with Digit
4. **num_123**
5. **num123**
6. **num@123** # contained the special Char
7. **While** # while is keyword but it is While (i.e. start with cpas)
8. **ELSE** # else keyword but is in upper case
9. **true** # it is Keyword
10. **NUMBER**

Data type is a special keyword used to allocate sufficient memory space for the data, in other words Data type is used for representing the data in main memory (RAM) of the computer.

In general every programming language is containing three categories of data types.
They are

- Fundamental or primitive data types
- Derived data types
- User defined data types.

Primary Data Type

Type	Size	Range	Default Value
byte	1 byte	-128 to 127	0
short	2 bytes	-32,768 to 32,767	0
int	4 bytes	-2,147,483,648 to 2,147,483, 647	0
long	8 bytes	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	0
float	4 bytes	3.4e-038 to 3.4e+038	0.0f
double	8 bytes	1.7e-308 to 1.7e+038	0.0d
char	2 byte	0 to 65,536 (unsigned)	'\u0000'
boolean	1 byte	true or false	false

Variable

- It is a data name used for storing a data value.
- Its value may be changed during the program execution.
- The value of variables keeps on changing during the execution of a program.

Declaration of Variable:

Sy: <data type> <variable name>

Ex: int num;

Initialization on variable:

Sy: <data type> <variable name>= <value>;

Ex: int num=34;

 char gen='m';

 float amt=444.5f;

Constant

- Values do not change during the execution of the program
- **Numerical constants:**
 - **Integer constants:** These are the sequence of numbers from 0 to 9 without decimal points or fractional part or any other symbols. Ex: 10, 20, + 30, - 14
 - **Real constants:** It is also known as floating point constants. Ex: 2.5, 5.342
- **Character constants:**
 - **Single character constants:** A character constant is a single character. Characters are also represented with a single digit or a single special symbol or white space enclosed within a pair of single quote marks.
Eg: 'a', '8', " ".
 - **String constants:** String constants are sequence of characters enclosed within double quote marks.
Eg: "Hello", "india", "444"

- Three permissible styles of comment in a Java technology program are:
 - `//` comment on one line
 - `/*` comment on one or more lines `*/`
 - `/**` documenting comment `*/`

- Which among the following encoding system can represent 65536 characters?
 - a. ASCII
 - b. Unicode
 - c. Both a and b
 - d. None of these
- Which among the following is not a Token?
 - a. Keywords
 - b. Literals
 - c. Identifiers
 - d. Data Type
- Which among the following is a keyword but not a literal?
 - a. for
 - b. null
 - c. true
 - d. false
- Which among the following is a valid float literal?
 - a. 12.36f
 - b. 12.36F
 - c. 12.36
 - d. Both a and b

➤ Which among the following is a valid octal integer literal?

a. 0178

b. 675

c. 0675

d. 0X675

➤ Which among the following is a valid method of initialising?

a. `boolean f=true;`

b. `boolean f=True;`

c. `boolean f='true';`

d. None of these

➤ Which among the following is not a punctuator?

a. ; semicolon

b. , comma

c. : colon

d. . Dot

➤ Which among the following is not a primitive data type?

a. int

b. float

c. String

d. char

- What is the largest possible value that can be stored in short data type?
 - a. $2^{15}-1$
 - b. $2^{31}-1$
 - c. 2^7-1
 - d. 2^63-1
- If a is of type int and b is of type float what would be the resultant data type of a+b?
 - a. Int
 - b. float
 - c. Double
 - d. short